

Figure 7. **Few-shot performance for generated image detection.** The bar chart reports *ACC* and *AP* under varying training data ratios (0.1%, 1%, 10%, 50%, 100%). Both metrics steadily improve as more training data becomes available.

5.4. Ablation Study

To evaluate the contributions of each component, we conduct an ablation study summarized in Table 1. Adding CIB yields a clear improvement over the baseline, increasing the average accuracy from 87.81% to 89.72% and the average AP from 87.32% to 99.38%, indicating that removing superfluous information strengthens task-relevant cues. Introducing AFCL leads to further gains, particularly in cross-model accuracy, which rises from 85.56% to 91.15%, showing that maintaining diverse and complementary cues improves robustness to distribution shifts. When all components are combined, the model achieves the best performance, reaching 92.81% average accuracy and 99.52% average AP. These results demonstrate that information purification and anti-collapse learning work synergistically to enhance both overall and cross-model generalization.

5.5. Few-shot Performance

To assess the data efficiency of our method, we train the detector using different proportions of the full dataset: 0.1% (320 samples), 1% (3.2k), 10% (32k), 50% (160k), and 100% (320k). As shown in Fig. 7, our method achieves strong performance even under limited supervision—for example, using only 0.1% of the data, it already obtains 80.98% *ACC* and 90.81% *AP*. As more data becomes available, both metrics improve steadily, reaching 92.81% *ACC* and 99.52% *AP* with the full dataset. These results demonstrate that our framework maintains high sample efficiency, achieving competitive performance even with limited data, and continues to improve as more training samples are provided, showing a strong ability to make effective use of supervision at all scales and to adapt to richer training signals.

5.6. Robustness to Unseen Perturbations

In real-world scenarios, images are affected by post-processing operations, making robustness to perturbations an essential aspect of evaluating detector reliability. We therefore assess performance under two commonly encountered distortions, JPEG compression and Gaussian blur, as shown in Fig. 8. As the perturbation strength increases, different detectors exhibit varying levels of sensitivity to de-

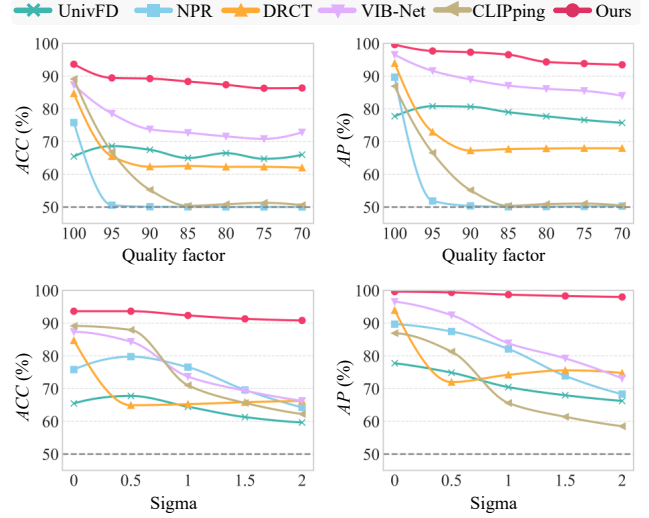


Figure 8. **Robustness evaluation under perturbations.** The top row shows performance under different JPEG compression quality factors, while the bottom row reports results under Gaussian blur with increasing sigma. Both *ACC* and *AP* are presented for each method, providing a comprehensive comparison of robustness against common post-processing distortions.

graded inputs. Methods that depend more heavily on localized or high-frequency artifacts tend to show larger decreases in performance, while approaches with stronger feature abstraction generally retain better stability. Overall, accuracy declines are observed to varying degrees across existing baselines as distortions intensify. Our method maintains the highest *ACC* and *AP* across all distortion levels in both settings. This robustness arises from cue purification and anti-collapse learning, which reduce over-reliance on fragile low-level artifacts that are easily disrupted by post-processing such as JPEG compression and Gaussian blur.

6. Conclusion

In this work, we revisit the core challenges of generalization in generated image detection and identify a fundamental limitation of existing detectors: during training, diverse forgery cues often collapse into a small set of dominant feature directions, leading to brittle decision boundaries and poor adaptation to unseen generative models. To address this issue, we introduce a unified detection framework that combines cue-level information purification, inter-cue diversity preservation, and class-specific authenticity modeling. Together, these components maintain heterogeneous and complementary sources of evidence, suppress spurious or redundant patterns, and construct a well-structured, authenticity-aware feature space. Extensive experiments across multiple benchmarks demonstrate that preserving diverse and high-rank representations is critical for achieving robust cross-model generalization.